SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA5611

COURSE OUTCOME COVERED

CO: 5 - *Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)*

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr. Tamil Selvi

QUESTIONS:Total Marks: Duration: 90 Minutes

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. Virtualization and Resource Management in Cloud RAN (BL5)
5. Edge AI for Intelligent 5G Network Optimization (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. Security and Privacy Challenges in Edge Computing for 5G (BL5)
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)
13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)

1. NB-IoT Architecture and Emerging Industrial Applications (BL5)
2. LTE-M Technology for Massive Machine-Type Communications (BL5)
3. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
4. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
5. Battery Life Optimization Strategies for Massive IoT Devices (BL5)
6. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
7. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
8. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
9. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
10. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
11. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
12. Digital Twin Technology for Smart City Infrastructure (BL5)
13. Edge AI Applications for Sustainable Smart Cities (BL5)
14. Industrial IoT Applications in Smart Manufacturing (BL5)
15. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
16. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
17. Private 5G Networks for Smart Factory Deployments (BL5)
18. Edge Computing for Real-Time Industrial Automation (BL5)
19. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
20. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
21. Intelligent Remote Patient Monitoring Using Wearable IoT Devices (BL5)
22. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
23. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)
24. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
25. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

Code of Conduct:

I _TEJHASH. P(192512495)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Tejhash. P

Signature of the student:

RUBRICS FOR CALCULATION:

Article Review					
Criteria	**Weightage**	**Level 4**	**Level 3**	**Level 2**	**Level 1**

		(Exemplary)	(Proficient)	(Developing)	(Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article’s purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

5G-BSS: 5G-Based Universal Blockchain Smart Sensors

The 5G-BSS (5G-Based Universal Blockchain Smart Sensor) framework introduces an architectural design for smart sensors, tailored specifically for high-security, ultra-low latency applications such as autonomous driving, intelligent transportation systems, and environmental monitoring. Smart sensors conventionally combine physical sensing components with microprocessors to capture, process, and transmit data. However, traditional deployments face critical bottlenecks when operating in high-demand environments. First, data transmission latency remains too high for real-time decision-making. Second, conventional communication channels lack adequate end-to-end encryption, exposing data to eavesdropping, signal interference, and integrity compromise. Third, relying on centralized cloud or edge storage leaves data vulnerable to hardware failures, unauthorized leaks, and tampering by malicious third parties or semi-trusted administrators.

To resolve these deficiencies, the 5G-BSS design synergizes three major technological capabilities directly onto the sensor platform. It integrates a dedicated 5G wireless communication module, such as the Huawei Baron MH5000-31, which utilizes high-bandwidth and ultra-reliable low-latency communication (URLLC) features to eliminate multi-tiered edge relay delays and upload sensor data instantaneously. For security, the system incorporates dedicated cryptographic algorithms and a USBKey peripheral, providing key management and hardware-isolated encryption to ensure strict data confidentiality and integrity prior to transmission. Furthermore, 5G-BSS embeds a lightweight blockchain client natively on a Raspberry Pi 4B microcomputer hardware platform, allowing the sensor to interact directly with a distributed ledger like FISCO BCOS. By committing cryptographically signed transaction data straight to an immutable blockchain, the sensor bypasses semi-trusted centralized storage completely.

Experimental evaluations conducted using JMeter on the FISCO BCOS blockchain framework demonstrate that 5G-BSS achieves an exceptional throughput of approximately 3,900 transactions per second (TPS) while maintaining low transaction latency between 400 and 450 milliseconds even under heavy operational workloads. By streamlining the data transmission path and executing encryption and hashing on-device, the architecture effectively guarantees data availability, confidentiality, and tamper-resistance. Ultimately, 5G-BSS offers a robust, scalable solution for mission-critical IoT networks where instant response times and uncompromised data authenticity are paramount.